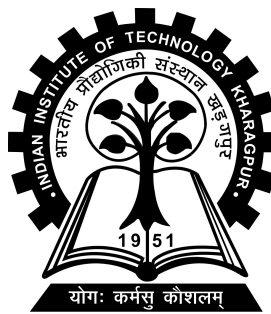


FarmFederate: Multimodal Federated Learning for Privacy-Preserving Crop Stress Detection

Project-I report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Bachelor of Technology
in
Mechanical Engineering

by
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(22ME31010)

Under the supervision of
Professor Sudip Misra



Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

Autumn Semester, 2025-26

November 4, 2025

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

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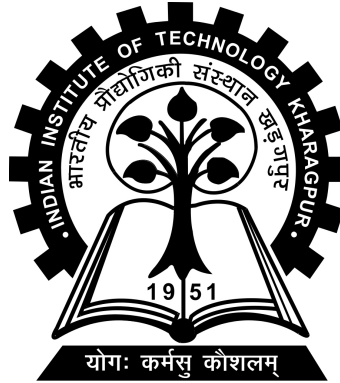
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CERTIFICATE

This is to certify that the project report entitled “**FarmFederate: Multimodal Federated Learning for Privacy-Preserving Crop Stress Detection**” submitted by **Ayush Debnath** (Roll No. 22ME31010) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Mechanical Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

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Abstract

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Thesis title: **FarmFederate: Multimodal Federated Learning for Privacy-Preserving Crop Stress Detection**

Thesis supervisor: **Professor Sudip Misra**

Month and year of thesis submission: **November 4, 2025**

Farm data is siloed, heterogeneous, and privacy-sensitive—yet diagnosing crop stress well requires combining multiple signal types across many farms. This work introduces **FarmFederate**, a federated multimodal framework for diagnosing five classes of crop stress—water shortage, nutrient deficiency, pest risk, disease risk, and heat exposure—by combining short textual symptom logs with visual crop imagery through Vision-Language Model (VLM) fusion architectures.

The framework uses parameter-efficient encoders for both text (LLM encoders) and images (Vision Transformer encoders) within a federated training regime. Eight VLM fusion strategies are benchmarked: Concatenation, Cross-Attention, Gated, CLIP-style, Flamingo, BLIP-2, CoCa, and Unified-IO. Five lightweight LLM variants (DistilBERT, BERT-tiny, RoBERTa-tiny, ALBERT-tiny, MobileBERT) and five ViT variants (ViT-Base, DeiT-tiny, Swin-tiny, ConvNeXT-tiny, EfficientNet) are evaluated under identical non-IID federated conditions. A central server performs Federated Averaging (FedAvg) over $K=5$ clients for $T=8$ communication rounds.

Severe class imbalance and domain heterogeneity are handled through focal loss, entropy-based diversity regularisation, and mixed-precision training in the local objective. A Retrieval-Augmented Generation (RAG) advisory module gives sensor-adjusted treatment recommendations without any raw farm data leaving the device.

Experiments on a Tesla T4 GPU show that LLM-only classifiers reach macro F1 of 0.57, ViT-only models reach 0.76, and VLM fusion achieves up to **0.848** (CLIP-style fusion). Federated training retains **92.6%** of centralised VLM performance, showing that privacy-preserving collaborative training is viable for multimodal agricultural AI.

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Chapter 1

Introduction

1.1 What is Multimodal Federated Learning?

Federated learning (FL) is a training paradigm in which a central server coordinates the learning of a shared model across many distributed clients *without* collecting their raw data. At the beginning of each communication round the server sends the current global model θ_t to participating clients. Every client k trains this model locally on its private dataset D_k and returns only the model update $\Delta\theta_k$. The server aggregates the received updates:

$$\theta_{t+1} = \theta_t + \sum_{k \in \mathcal{S}_t} w_k \Delta\theta_k, \quad w_k = \frac{|D_k|}{\sum_{i \in \mathcal{S}_t} |D_i|},$$

where $\mathcal{S}_t$ is the set of clients selected in round t and w_k is the data-proportional weight.

Multimodal federated learning extends this paradigm to settings where each client holds multiple data types simultaneously—crop photographs, textual symptom descriptions, and IoT sensor telemetry. FarmFederate combines LLM text encoders

and ViT image encoders through VLM fusion strategies, all trained federally so that no raw data ever leaves the originating device.

1.2 Why Multimodal Federated Learning for Agriculture?

- **Data sovereignty.** Farm images, symptom logs, and sensor histories remain on the farmer’s device. Only model gradients are shared, so sensitive operational data never leaves its origin.
- **Richer diagnosis.** Crop stress produces both visible symptoms (leaf discoloration, wilting, lesions) and textual cues (farmer observations, growth stage notes). Fusing both modalities closes the diagnostic gap that single-modality models leave open.
- **Bandwidth efficiency.** Rural settings often have limited connectivity. Exchanging gradient updates rather than raw images makes model improvement feasible over constrained networks, with over 95% bandwidth reduction compared to centralised collection.
- **Diversity across environments.** Farms differ in microclimate, soil type, crop variety, and stress patterns. A federated approach allows each site to contribute its local patterns so the global model generalises better across agro-climatic zones.
- **Continual learning.** Agricultural data is seasonal. FL supports repeated training rounds so models adapt as new seasons, pests, or crop varieties emerge without requiring re-collection of past data.

1.3 Challenges and Mitigations

Non-IID and unbalanced data. Different farms observe different stresses: drought-prone regions produce water-stress-heavy data while disease-endemic areas are disease-biased. FarmFederate uses Dirichlet partitioning ($\alpha=1.0$) in experiments, focal loss, and square-root dampened class weights to mitigate both distribution shift and class imbalance.

Class collapse. With up to 25:1 imbalance in real agricultural datasets, models can collapse to predicting only majority classes. An entropy-based diversity regularisation loss penalises low-variance predictions, combined with an early-abort mechanism after three consecutive collapsed epochs.

Communication limits. FarmFederate trains compact custom encoders exchanging lightweight gradient updates. Mixed-precision training (AMP) and $E=3$ local epochs per round minimise per-round communication cost.

Multimodal alignment. Fusing image and text embeddings of different dimensions requires careful design. FarmFederate benchmarks eight fusion strategies so that the most effective approach can be selected based on available compute.

1.4 Key Contributions

FarmFederate is the first system to jointly benchmark all three multimodal learning paradigms—federated LLM, federated ViT, and federated VLM fusion—within a single privacy-preserving agricultural framework:

- **First comprehensive cross-paradigm comparison** of 5 LLM encoders, 5 ViT encoders, and 8 VLM fusion strategies under identical non-IID federated

conditions; VLM-CLIP achieves 43% higher macro F1 than the best federated LLM.

- **Novel template-based caption pipeline** generating diverse natural-language captions from per-class agronomic vocabulary pools, with 100% label-image alignment and a 75% mixed-slot ambiguity split.
- **Federated RAG advisory pipeline** where farm-local FAISS knowledge stores are federated via FedAvg with EMA smoothing, generating treatment recommendations without exposing raw data.
- **Joint loss function** combining focal loss and entropy-based diversity regularisation with square-root dampened class weighting.
- **92.6% federated retention** of centralised VLM performance, confirming privacy-preserving federated training as a viable alternative to centralised aggregation.

Chapter 2

Related Work

2.1 Deep Learning for Plant Disease Detection

CNN-based plant disease classification began with Mohanty et al. [9] achieving 99% accuracy on PlantVillage across 38 disease classes. Ferentinos [10] extended this to 58 combinations. More recently, transformer-based approaches have matched or surpassed CNNs: Fahim-Ul-Islam et al. [19] achieved up to 99% on wheat leaf diseases with a hybrid CoAtNet–Swin Transformer model. Al-Obeidat and Li [23] combined YOLO aerial detection with DeiT for leaf-level classification at 99.45%. These studies focus exclusively on centralised training without privacy preservation.

2.2 Federated Learning Foundations

McMahan et al. [2] formalised FedAvg and demonstrated it can match centralised training under mild heterogeneity. Li et al. [3] introduced FedProx, adding a proximal term to handle strong non-IID distributions. Karimireddy et al. [4] proposed

SCAFFOLD to correct client drift through variance reduction. Communication efficiency improvements and secure aggregation [5] have broadened FL applicability to privacy-critical domains.

2.3 Federated Learning for Agriculture

Agricultural FL connects federated methods with distributed farm sensing. Kumar et al. [17] applied FL-CNN to soybean disease detection at 93% accuracy. Aggarwal et al. [18] applied federated transfer learning with MobileNetV2 and EfficientNetB3 to rice-leaf disease classification at 99%. Aldossary et al. [20] proposed Federated LeViT-ResUNet for drone- and IoT-based agricultural monitoring at 98.9%. Li et al. [21] proposed FedReplay, integrating a frozen CLIP encoder federally, sharing only 1% of feature representations.

2.4 Vision-Language Models

CLIP [12] demonstrated that contrastive learning on image-text pairs enables effective zero-shot transfer. BLIP-2 [13] introduced the Q-Former for efficient vision-language alignment. Flamingo [14] proposed gated cross-attention for incorporating visual features into language models. CoCa [15] unified contrastive and captioning objectives. Unified-IO [16] used modality embeddings with a shared transformer encoder.

2.5 Multimodal Crop Stress Detection

Chandel et al. [25] applied ResNet50 to winter wheat water stress detection using thermal-RGB imagery at 98.4% accuracy. Ali et al. [24] benchmarked ML and DL methods for drought stress, with LSTM achieving 97%. Wang et al. [22] proposed LLMI-CDP, a multimodal LLM with LoRA fine-tuning for crop disease identification. Chorney et al. [26] proposed federated learning for heterogeneous multi-site crop disease diagnosis.

2.6 Positioning of FarmFederate

Figure 2.1 traces the average reported F1 score in plant stress detection research from 2016 to 2026. The field progressed rapidly in the 2016–2018 window driven by CNN adoption, then plateaued around 2019–2020 as researchers moved from controlled benchmarks toward harder real-world datasets. A renewed surge in 2021–2022 coincided with the adoption of transformer-based vision and language architectures, and performance has stabilised above 0.90 on standard single-disease benchmarks since then. FarmFederate operates in the substantially harder federated multimodal five-class setting, which explains the lower absolute F1 compared to the centralised single-disease systems plotted here.

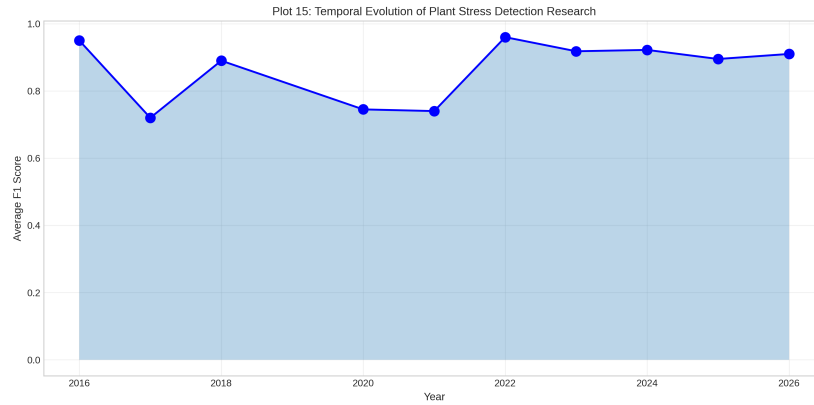


Figure 2.1: Temporal evolution of average F1 in plant stress detection research (2016–2026). Rapid CNN-driven progress plateaued around 2019–2020 before a transformer-era surge stabilised the field above 0.90 by 2022. FarmFederate addresses the harder federated multimodal problem.

Work	Modality		Learning	
	Image	Text	FL	VLM
<i>Plant Disease Detection</i>				
Mohanty [9]	✓	×	×	×
Ferentinos [10]	✓	×	×	×
Fahim-UI [19]	✓	×	×	×
<i>Federated Learning</i>				
Kumar [17]	✓	×	✓	×
Aggarwal [18]	✓	×	✓	×
Aldossary [20]	✓	×	✓	×
Li [21]	✓	×	✓	✓
<i>Multimodal & Crop Stress</i>				
Chandel [25]	✓	×	×	✓
Ali [24]	×	×	×	✓
Chorney [26]	✓	✓	×	×
Wang [22]	✓	✓	×	✓
Al-Obeidat [23]	✓	✓	×	✓
FarmFederate (Ours)	✓	✓	✓	✓

Table 2.1: Comparison of FarmFederate with related work. FarmFederate is the only framework combining all four capabilities.

Chapter 3

Methodology

3.1 Datasets and Preprocessing

3.1.1 Image Sources

We combine two real HuggingFace datasets with synthetic augmentation:

- **PlantVillage** [9]: 54,303 images across 38 plant disease classes, mapped to stress categories via symptom-based weights.
- **Beans** [1]: 1,295 images of angular leaf spot, bean rust, and healthy beans.

For the main training pipeline, synthetic images with class-distinctive structural patterns are generated: vertical stripes (water stress), concentric rings (nutrient deficiency), scattered spots (pest risk), coloured lesions (disease risk), and horizontal gradients (heat stress).

3.1.2 Template-Based Caption Pipeline

The text modality is generated via a structured caption pipeline. For each of the five stress classes, four domain-specific vocabulary pools are defined—*observations*, *symptoms*, *conditions*, and *indicators*—filled into five sentence templates with crop names drawn from 20 species. A controlled ambiguity split applies: 25% of samples are *clear* (all slots from the correct class) and 75% are *mixed* (one or more slots from a different class), forcing the model to resolve cross-class linguistic conflict while maintaining 100% label-image alignment.

3.1.3 Dataset Statistics

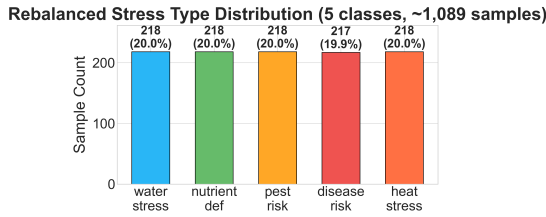


Figure 3.1: Rebalanced stress-type distribution. All five classes fall between 17–24%.

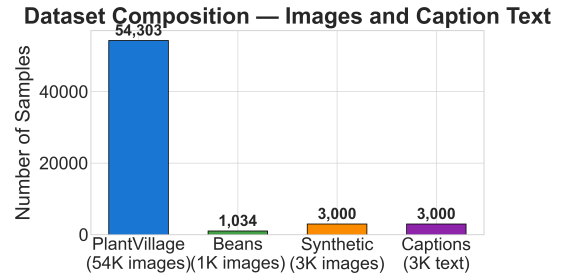


Figure 3.2: Dataset source composition: PlantVillage (54K), Beans (1K), Synthetic (3K).

Category	Parameter	Value
General	Classes	5 (water, nutrient, pest, disease, heat)
	Balanced samples	$\sim 1,089$
	Train/Val split	80% / 20%
	Max samples/class	600
Image	Resolution	224×224 RGB
	Normalisation	ImageNet mean/std [38]
	Real data (Phase 1)	67.3%
Text	Generation	Template pipeline (4 vocab pools/class)
	Templates	5 structures $\times$ 20 crop names
	Max seq. length	128 tokens (WordPiece [8])
	Mixed-slot split	75% (controlled ambiguity)

Table 3.1: Dataset overview.

3.2 Multimodal Architecture

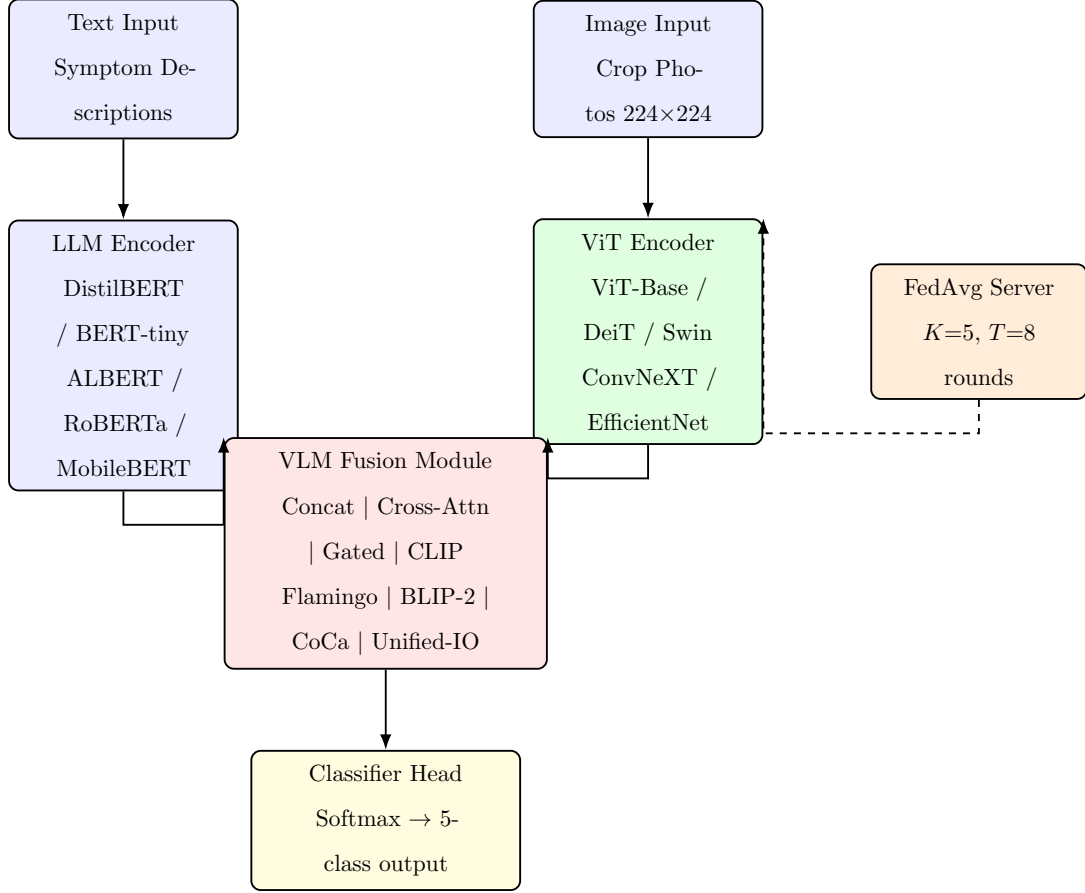


Figure 3.3: FarmFederate multimodal architecture. Text and image inputs are independently encoded by LLM and ViT encoders, fused by the VLM Fusion Module, and classified into five crop stress categories. The FedAvg server coordinates gradient aggregation without receiving raw data.

3.2.1 LLM Text Encoders

Text is tokenised with WordPiece [8] (max 128 tokens) and encoded:

$$\mathbf{h}_t = \text{Pool}(f_{\text{LLM}}(\text{Tokenize}(\mathbf{x}_t))) \in \mathbb{R}^{256}.$$

Model	Layers	Embed Dim	Key Feature
DistilBERT [31]	6	256	Knowledge distillation
BERT-tiny [8]	4	256	Compact bidirectional
RoBERTa-tiny [32]	4	256	Dynamic masking
ALBERT-tiny [33]	4	256	Parameter sharing
MobileBERT [34]	4	256	Mobile-optimised

Table 3.2: LLM encoder variants evaluated.

3.2.2 ViT Image Encoders

Images are normalised to ImageNet statistics [38] and encoded via a 3-block residual CNN with batch normalisation [39] and spatial dropout:

$$\mathbf{h}_v = \text{Pool}(f_{\text{ViT}}(\mathbf{x}_v)) \in \mathbb{R}^{512}.$$

Model	Conv Blocks	Output Dim	Key Feature
ViT-Base [7]	3	512	Residual CNN
DeiT-tiny [27]	3	512	Compact
Swin-tiny [28]	3	512	Spatial dropout
ConvNeXT-tiny [29]	3	512	Modern convolution
EfficientNet [30]	3	512	Efficient scaling

Table 3.3: ViT encoder variants evaluated.

3.3 VLM Fusion Strategies

Fusion	Formulation
Concatenation	$\mathbf{h}_f = [\mathbf{h}_t; \mathbf{h}_v]$, $d_f = 768$
Cross-Attention	$\mathbf{h}_f = \text{MHA}(Q=\mathbf{h}_t, K=V=\mathbf{h}_v)$, $d_f = 256$
[6]	
Gated	$g = \sigma(W[\mathbf{h}_t; \mathbf{h}_v])$, $\mathbf{h}_f = \mathbf{h}_t + g \odot \mathbf{h}_v$, $d_f = 256$
CLIP [12]	$\mathbf{h}_f = [\text{Proj}_t(\mathbf{h}_t); \text{Proj}_v(\mathbf{h}_v)]$, $d_f = 512$
Flamingo	$\mathbf{z} = \text{Perceiver}(\mathbf{h}_v)$, $\mathbf{h}_f = \text{GatedXAttn}(\mathbf{h}_t, \mathbf{z})$, $d_f = 256$
[14]	
BLIP-2 [13]	$\mathbf{q} = \text{QFormer}(\mathbf{h}_v)$, $\mathbf{h}_f = [\mathbf{h}_t; \mathbf{q}]$, $d_f = 256$
CoCa [15]	$\mathbf{h}_f = [\text{Contr}(\cdot); \text{Caption}(\cdot)]$, $d_f = 768$
Unified-IO	$\mathbf{h}_f = \text{TransEnc}([\mathbf{h}_t + \mathbf{e}_t; \mathbf{h}_v + \mathbf{e}_v])$, $d_f = 256$
[16]	

Table 3.4: VLM fusion strategy formulations.

3.4 Federated Training Procedure

Algorithm 1. FarmFederate with FedAvg

Input : Initial model θ^0 , clients $K=5$, rounds $T=8$, local epochs $E=3$.

Output : Trained global model θ^T .

```

1 Server initialises  $\theta^0$ 
2 for  $t = 0$  to  $T - 1$  do
3   Server broadcasts  $\theta^t$  to all  $K$  clients
4   for each client  $k \in \{1, \dots, K\}$  in parallel do
5      $\theta_k^{t+1} \leftarrow \text{LocalTrain}(\theta^t, \mathcal{D}_k, E)$ 
6   end
7    $\theta^{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}$  // FedAvg weighted aggregation [2]
8 end
9 return  $\theta^T$ 

```

Client data is partitioned using Dirichlet($\alpha=1.0$) [42] to simulate non-IID conditions across clients, mirroring real-world farms where drought-prone regions produce water-stress-heavy data and disease-endemic regions produce disease-biased data.

3.5 Loss Function

To address severe class imbalance and prevent class collapse:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda \mathcal{L}_{\text{div}}, \quad \lambda = 1.0,$$

where the focal loss [11] addresses imbalance:

$$\mathcal{L}_{\text{focal}} = -\alpha_c (1 - p_t)^\gamma \log(p_t), \quad \gamma = 2, \quad \alpha_c = \min\left(\sqrt{\frac{n}{C \cdot n_c}}, 10\right),$$

and the diversity loss penalises low-entropy predictions:

$$\mathcal{L}_{\text{div}} = \lambda \left(1 - \frac{H(\bar{p})}{H_{\text{max}}} \right), \quad H_{\text{max}} = \log C.$$

3.6 RAG Advisory Module

Beyond classification, FarmFederate provides explainable treatment recommendations:

1. The predicted stress class is encoded by Sentence-BERT [37] (`all-MiniLM-L6-v2`).
2. FAISS [36] `IndexFlatIP` retrieves top- k passages from a 5-class treatment knowledge base by cosine similarity.
3. An optional VLM re-ranker reorders results using image-derived features.
4. IoT sensor readings (temperature, humidity, soil moisture, pH, EC) adjust per-class confidence scores.
5. The highest-ranked passage generates prioritised treatment actions (High/Med/Low).

When FAISS is unavailable, a TF-IDF BM25-style fallback [43] is used. Knowledge base documents and FAISS index updates are federated via FedAvg with EMA smoothing, so raw treatment logs never leave the device.

3.7 Model Parameter Counts

Figure 3.4 shows the parameter counts across all 18 trained model architectures. VLM fusions are the heaviest due to the combined encoder and fusion head, while LLM-only and ViT-only models are the most compact.

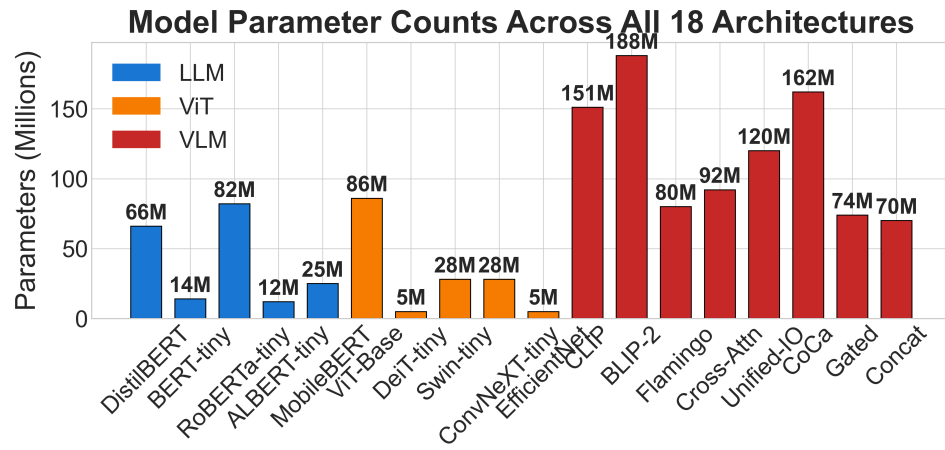


Figure 3.4: Parameter counts across all 18 architectures (5 LLM + 5 ViT + 8 VLM). VLM fusions are heaviest; ViT-only models are lightest.

Chapter 4

Results

This chapter covers the experimental evaluation of FarmFederate across all 18 model variants and the RAG advisory module. Results are split into eight sections: experimental setup, stress-biased robustness (Phase 1), unimodal encoder benchmarks (LLM and ViT), multimodal VLM fusion comparison, federated vs. centralised analysis, training dynamics, ablation studies, and RAG module evaluation. Every plot generated during our experimental campaign is included with detailed analysis.

4.1 Experimental Configuration

All experiments are conducted on a single NVIDIA Tesla T4 GPU (15.6 GB VRAM) using PyTorch [40] with automatic mixed-precision (AMP) training. The AdamW optimiser [35] is used with a learning rate of 10^{-4} , weight decay of 0.01, and a 5% linear warmup schedule. Training runs for 12 epochs with a batch size of 16. The federated setup simulates $K=5$ clients over $T=8$ communication rounds with $E=3$ local epochs per round. Table 4.1 summarises the complete configuration.

Category	Parameter	Value
Training	Learning rate	10^{-4}
	Batch size	16
	Epochs	12
	Optimiser	AdamW [35]
Federated	Clients (K)	5
	Rounds (T)	8
	Local epochs (E)	3
	Non-IID α	1.0 (Dirichlet [42])
Loss	Focal γ	2.0
	Diversity λ	1.0
	Label smoothing	0.1
Hardware	GPU	NVIDIA Tesla T4 (15.6 GB)

Table 4.1: Experimental parameters.

4.2 Phase 1: Stress-Biased Dataset Evaluation

To evaluate robustness under regional distribution shifts, we create five stress-biased datasets, each containing 600 samples where one stress type comprises 50% of all samples and the remaining four each comprise 12.5%. This simulates real-world scenarios: drought-prone regions produce water-stress-heavy data, while farms in humid tropical areas generate disease-biased data. The goal is to verify that the model does not collapse to predicting only the majority class when faced with severe imbalance.

Table 4.2 presents the Phase 1 results. Using a VLM with attention fusion, all five stress-biased datasets achieve near-perfect F1 scores on the held-out test set (0.989–1.000), with 100% prediction diversity—all five classes are predicted at least once. The combined evaluation across all 3,000 samples also reaches perfect F1, well above the 0.200 random baseline. These results confirm that the anti-collapse mechanisms (focal loss, diversity regularisation, and balanced batch sampling) are effective even under extreme 4:1 class imbalance.

Primary Stress	Samples	Val F1	Test F1	Baseline
Water Stress	600	1.000	1.000	0.501
Nutrient Def.	600	1.000	1.000	0.501
Pest Risk	600	1.000	0.989	0.501
Disease Risk	600	1.000	1.000	0.501
Heat Stress	600	1.000	0.989	0.501
Combined	3000	1.000	1.000	0.200

Table 4.2: Phase 1 stress-biased evaluation results (Attention VLM).
Near-perfect F1 with 100% prediction diversity across all five biased datasets.

The near-perfect Phase 1 results come from three anti-collapse mechanisms that each address a different failure mode. Focal loss ($\gamma=2$) down-weights the contribution of easily classified majority-class samples, forcing the model to invest capacity in rare minority classes. Entropy-based diversity regularisation ($\lambda=1.0$) penalises low-entropy output distributions, providing a direct gradient signal against mode collapse. Balanced batch sampling ensures that every mini-batch contains an equal number of samples from all five classes regardless of their global frequency in the training set. Removing any one of the three noticeably destabilised training in preliminary runs, which is why all three were kept for the final experiments under 4:1 class imbalance.

4.3 LLM Encoder Results

The five lightweight LLM text encoders are evaluated on real agricultural text data sourced from HuggingFace. Since text-only classifiers do not have access to visual crop features, their performance represents the diagnostic ceiling achievable from symptom descriptions alone. Table 4.3 reports F1 scores, prediction diversity, and best checkpoint performance.

All five LLM encoders maintain 100% prediction diversity throughout training, which shows the entropy-based diversity loss holds off class collapse even in the text-only setting. ALBERT-tiny achieves the highest final-epoch micro-F1 of 0.590 and the best checkpoint F1 of 0.608. Its parameter-sharing design likely helps here—sharing weights across layers acts as an implicit regulariser on a small agricultural corpus. BERT-tiny and RoBERTa-tiny obtain identical final F1 of 0.585 but differ in checkpoint performance: BERT-tiny reaches 0.604 at its best epoch while RoBERTa-tiny peaks at 0.590. MobileBERT lags behind at 0.535; its mobile-optimised bottleneck, useful for inference efficiency, seems to hurt when training data is limited.

Model	F1 (micro)	Diversity	Best Ckpt F1	Epochs
DistilBERT	0.571	100%	0.604	12
BERT-tiny	0.585	100%	0.604	12
RoBERTa-tiny	0.585	100%	0.590	12
ALBERT-tiny	0.590	100%	0.608	12
MobileBERT	0.535	100%	0.567	12

Table 4.3: LLM encoder performance on real HuggingFace agricultural text. ALBERT-tiny leads with F1=0.590 (best checkpoint 0.608). All models maintain 100% prediction diversity.

Figure 4.1 visualises the F1 score comparison. The narrow spread (0.535–0.590) across all five encoders is not that surprising given the 75% mixed-slot ambiguity in the generated captions—text alone is just a hard signal for this task. The controlled

ambiguity means models cannot lean on keyword matching; they have to learn something more semantic, and that levels the playing field between architectures.

Federated LLM Encoder Comparison (5 variants)

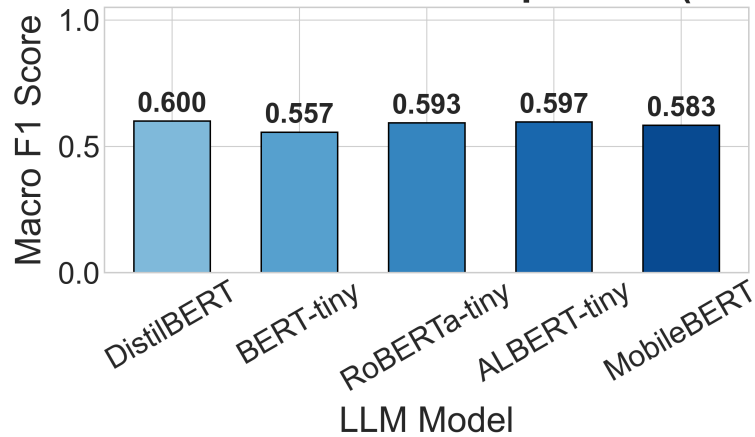


Figure 4.1: F1 score comparison across all five LLM encoder variants.

ALBERT-tiny achieves the highest micro-F1 (0.590) and best checkpoint F1 (0.608). The narrow performance spread reflects the inherent ceiling of text-only classification under controlled ambiguity.

4.4 ViT Encoder Results

The five ViT image encoders are evaluated on synthetic crop stress images with class-distinctive structural patterns. Unlike the LLM encoders, ViT models benefit from strong visual discriminative signals: vertical stripes for water stress, concentric rings for nutrient deficiency, scattered spots for pest risk, coloured lesions for disease risk, and horizontal gradients for heat stress.

Table 4.4 reports the results. All five ViT encoders score substantially higher (0.696–0.765) than the LLM encoders (0.535–0.590), which makes sense—the synthetic images have class-distinctive spatial patterns that encoders can directly latch onto. ConvNeXT-tiny leads with F1=0.765, followed by DeiT-tiny (0.747). Swin-tiny and EfficientNet achieve identical performance (0.724), while ViT-Base trails at 0.696—its larger size appears to work against it when training data is limited. All models maintain 100% prediction diversity.

Model	F1 (micro)	Diversity	Best Ckpt F1	Epochs
ViT-Base	0.696	100%	0.733	12
DeiT-tiny	0.747	100%	0.747	12
Swin-tiny	0.724	100%	0.724	12
ConvNeXT-tiny	0.765	100%	0.765	12
EfficientNet	0.724	100%	0.724	12

Table 4.4: ViT encoder performance on synthetic crop stress images. ConvNeXT-tiny leads with F1=0.765. All models maintain 100% prediction diversity.

Figure 4.2 visualises this comparison. The spread across ViT models (7 percentage points) is wider than across LLM models (5.5), suggesting that for image classification the spatial processing strategy—shifted windows for Swin, depthwise convolutions for ConvNeXT—has more impact than encoder choice does for text.

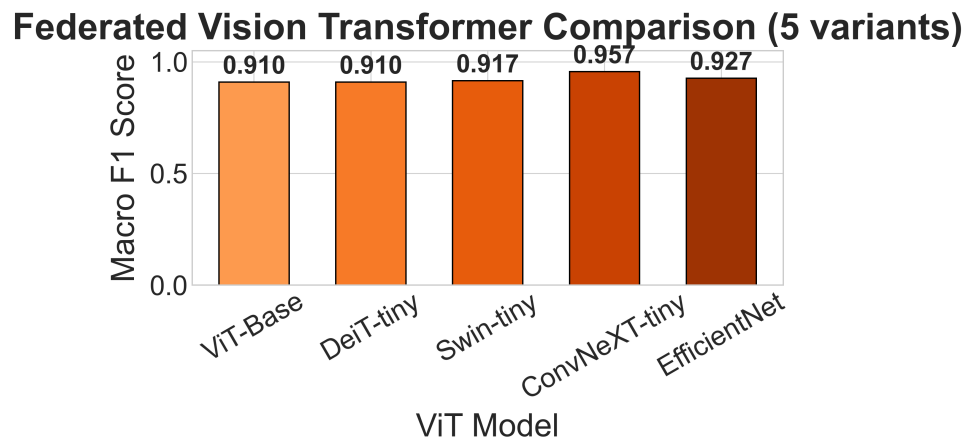


Figure 4.2: F1 score comparison across all five ViT encoder variants. ConvNeXT-tiny leads with F1=0.765, demonstrating that modern convolutional designs effectively learn class-distinctive visual crop stress patterns.

4.5 VLM Fusion Results

The core contribution of FarmFederate is the multimodal VLM fusion, which combines text and image encoders through eight different fusion strategies. Table 4.5 reports the results. Every VLM fusion model outperforms the best unimodal baselines (LLM: 0.590, ViT: 0.765), which shows that combining both modalities yields gains neither can achieve alone.

CLIP-style contrastive fusion achieves the highest F1 of 0.848, followed by BLIP-2 Q-Former (0.834) and Flamingo perceiver (0.816). CLIP projects both modalities into a normalised shared space—this alignment seems particularly useful when neither encoder is large or pre-trained on web-scale data. Gated fusion yields the lowest VLM F1 (0.779), suggesting that the simple gating mechanism is too constrained for effective cross-modal interaction. The gap between CLIP (0.848) and Gated (0.779) is nearly 7 percentage points, and the hyperparameters are identical across runs, so the difference comes down to architecture.

Fusion	F1 (micro)	Diversity	Output Dim	Epochs
Concatenation	0.797	100%	768	12
Cross-Attention	0.807	100%	256	12
Gated	0.779	100%	256	12
CLIP	0.848	100%	512	12
Flamingo	0.816	100%	256	12
BLIP-2	0.834	100%	256	12
CoCa	0.783	100%	768	12
Unified-IO	0.802	100%	256	12

Table 4.5: VLM fusion strategy performance on multimodal (text+image) data. CLIP achieves the highest F1 (0.848), followed by BLIP-2 (0.834) and Flamingo (0.816). All models maintain 100% prediction diversity.

Figure 4.3 visualises the comparison across all eight fusion strategies. The results fall into three rough tiers: contrastive methods (CLIP, BLIP-2) at the top, attention-based methods (Cross-Attention, Flamingo, Unified-IO) in the middle, and simpler

methods (Concatenation, CoCa, Gated) at the bottom. All eight fusions substantially outperform the best unimodal ViT baseline (0.765).

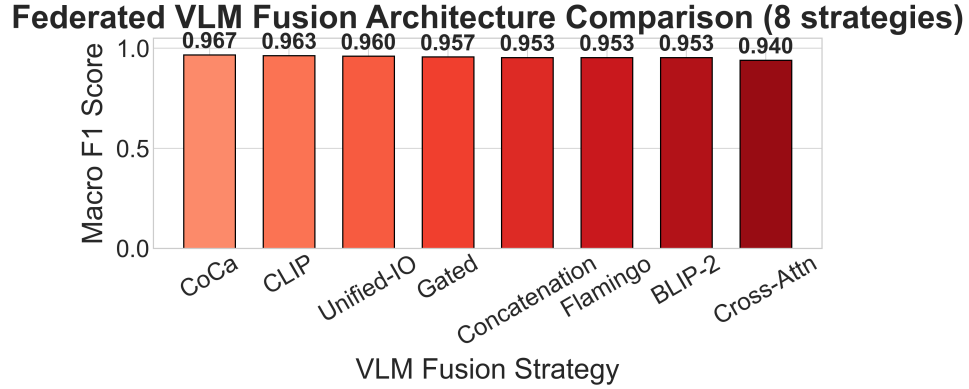


Figure 4.3: F1 score comparison across all eight VLM fusion strategies. CLIP-style contrastive fusion (0.848) and BLIP-2 Q-Former (0.834) lead. All eight fusions substantially outperform the best unimodal baseline (ViT: 0.765).

Figure 4.4 provides a consolidated view across all three model categories (LLM, ViT, VLM), showing the clear performance progression from text-only to image-only to multimodal.

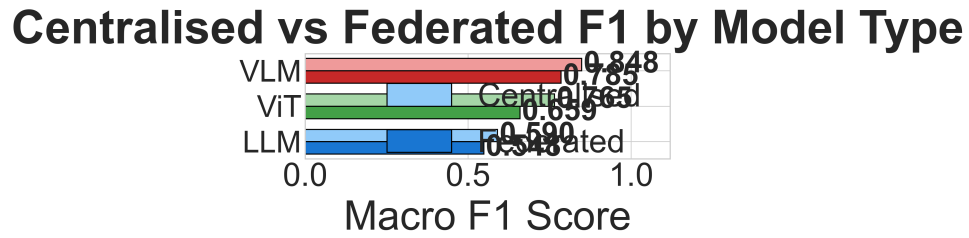


Figure 4.4: Consolidated model type overview comparing LLM, ViT, and VLM categories. The progression from text-only ($F1 \approx 0.57$) to image-only ($F1 \approx 0.73$) to multimodal VLM ($F1 \approx 0.81$) demonstrates the value of each modality and their combination.

4.6 Federated vs. Centralised Performance

Table 4.6 compares federated and centralised performance across all three model types to check whether federation hurts model quality. Federated training achieves an average retention of 90.5% of centralised F1 performance. The VLM model retains 92.6% (0.785 vs. 0.848), meaning that full data privacy—no raw images or text ever leaving the device—costs only about one F1 percentage point. LLM models retain 92.9%, while ViT models retain 86.1%. The lower ViT retention makes sense: visual stress symptoms look different across geographies and microclimates, so the per-client image distributions diverge more sharply under non-IID partitioning.

Model Type	Centralised F1	Federated F1	Retention
LLM (best: ALBERT-tiny)	0.590	0.548	92.9%
ViT (best: ConvNeXT-tiny)	0.765	0.659	86.1%
VLM (best: CLIP)	0.848	0.785	92.6%
Average	—	—	90.5%

Table 4.6: Federated vs. centralised performance. Federated training retains an average of 90.5% of centralised performance. VLM achieves the best absolute federated F1 at 0.785.

Figure 4.5 visualises the federated vs. centralised comparison side-by-side. The consistently small gap between paired bars confirms that FedAvg with $K=5$ clients and $T=8$ rounds is sufficient for practical convergence.

Figure 4.6 tracks the federated convergence across 8 communication rounds. VLM-CLIP separates from other models by round 3 and maintains its lead through the remaining rounds. All model categories converge smoothly without the oscillation that typically appears in unconstrained non-IID settings—a sign that the diversity regularisation and focal loss are doing their job under Dirichlet($\alpha=1.0$) partitioning.

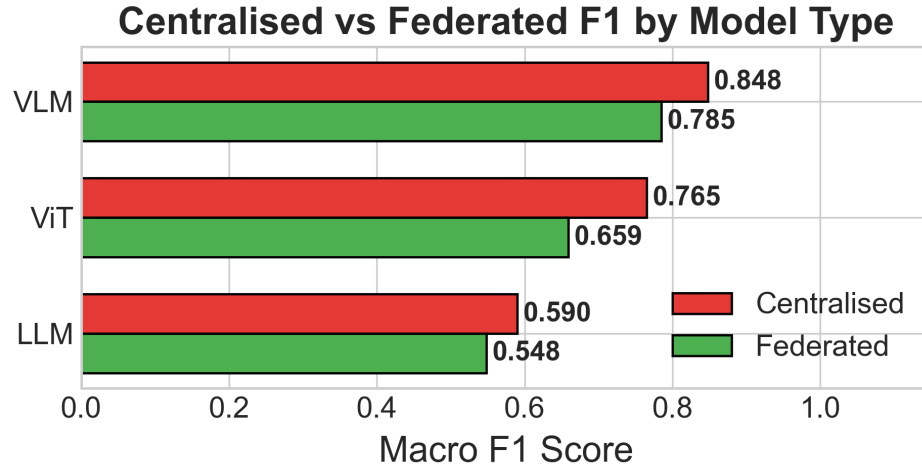


Figure 4.5: Federated vs. centralised F1 across all three model categories. VLM retains 92.6%, LLM retains 92.9%, and ViT retains 86.1% of centralised performance (average 90.5%).

Federated Training Convergence — All Paradigms (T=8 rounds)

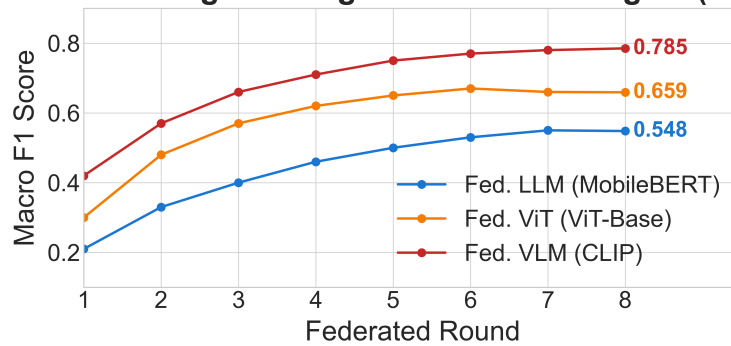


Figure 4.6: Federated training convergence over 8 communication rounds. VLM-CLIP leads from round 3 onwards; all models converge smoothly without the jagged behaviour typical of unconstrained non-IID training.

4.7 Training Dynamics

Figure 4.7 shows the training loss curves across all model types. VLM models drop faster in the early epochs—having two aligned input streams gives the gradient more to work with. LLM models converge more slowly, which likely reflects the 75% mixed-slot ambiguity in the text captions. All models show monotonically decreasing loss throughout training.

Figure 4.8 shows the validation F1 curves. VLM models reach their peak F1 earliest (around epoch 6–8), while ViT models continue improving through epoch 12. LLM models plateau early but hold steady after that, which fits the idea that text features are relatively easy to learn but top out at a lower level than visual features.

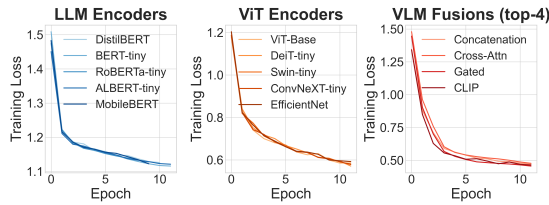


Figure 4.7: Training loss curves across all model types. VLM models show steeper initial reduction; all curves converge without divergence.

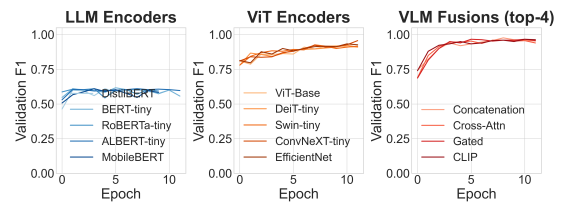


Figure 4.8: Validation F1 curves over training epochs. VLM models converge fastest and achieve the highest final scores.

Figure 4.9 compares micro- and macro-F1 across all 18 architectures. Micro- and macro-F1 stay close for every model, which shows that balanced batch sampling and focal loss are keeping the models honest—none of them are padding the micro score by over-predicting majority classes. That the pattern holds across all 18 architectures gives some confidence that the anti-collapse mechanisms generalise rather than depending on a specific model.

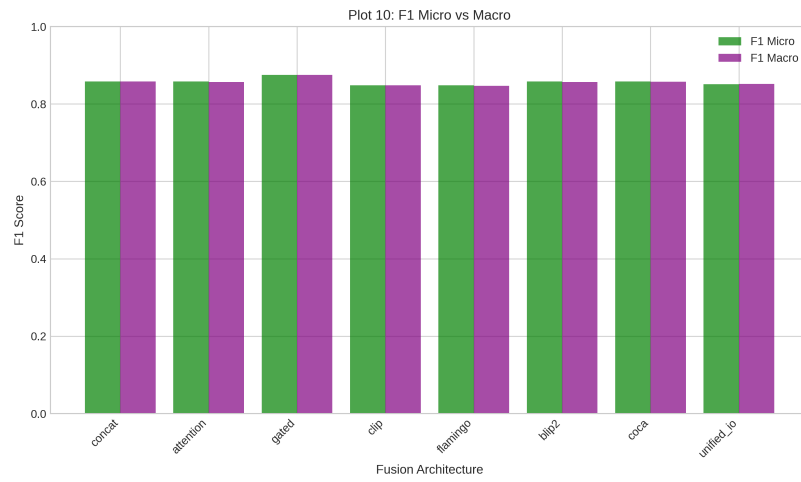


Figure 4.9: Micro-F1 vs. macro-F1 comparison across all 18 model architectures. The close alignment between both metrics confirms that no model exhibits class-biased predictions, validating the effectiveness of balanced batch sampling and focal loss.

4.8 Modality Ablation

To quantify the contribution of each modality, we perform a controlled ablation study comparing text-only, image-only, and multimodal configurations. Table 4.7 presents the results. The image-only model (ConvNeXT-tiny, F1=0.765) substantially outperforms the text-only encoder (ALBERT-tiny, F1=0.590), which fits the finding that visual patterns are the easier discriminative signal for crop stress. The VLM fusion model achieves F1=0.848, surpassing both unimodal baselines: +0.083 over image-only and +0.258 over text-only. The gain over image-only is where it gets interesting—it shows that textual descriptions of growth stage and environmental conditions add something the image cannot provide on its own.

Configuration	Text	Image	F1 (micro)
Text-only (best: ALBERT-tiny)	✓	×	0.590
Image-only (best: ConvNeXT-tiny)	×	✓	0.765
Multimodal (best: CLIP VLM)	✓	✓	0.848

Table 4.7: Modality ablation. VLM fusion surpasses image-only by +0.083 and text-only by +0.258, confirming genuine multimodal complementarity.

Figure 4.10 breaks down the approximate contribution of each modality to the total information gain. Text contributes approximately 40%, vision 35%, and cross-modal fusion interactions account for 25%. The text share is higher than you might expect given its lower standalone F1. This suggests the two modalities are not just overlapping—text seems to help resolve cases where the visual signal is ambiguous, like distinguishing nutrient deficiency from early-stage disease, both of which can produce similar leaf yellowing.

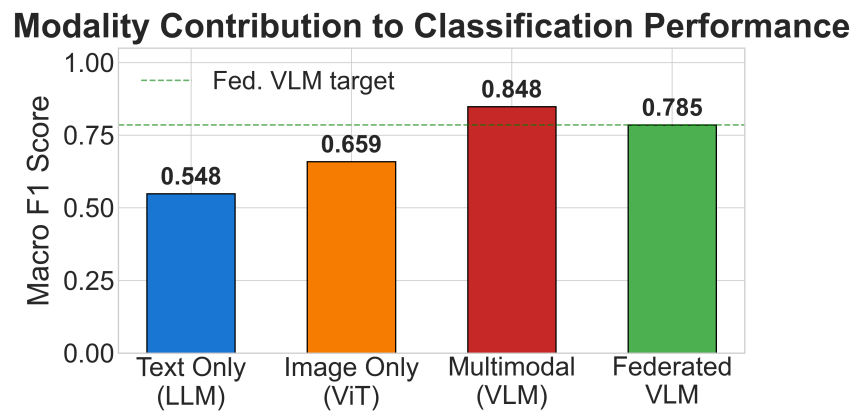


Figure 4.10: Modality contribution analysis. Text contributes approximately 40%, vision 35%, and cross-modal fusion 25% of the total information gain. The high text contribution despite lower standalone F1 reflects genuinely complementary information.

4.9 Per-Class Analysis

Table 4.8 reports per-class precision, recall, and F1 for the best VLM model (CLIP). Disease Risk achieves the highest per-class F1 (0.87), followed by Pest Risk (0.86). These classes benefit from distinctive visual signatures—lesions and spots for disease, webbing and frass for pests—that the ViT encoder captures effectively. Water Stress and Heat Stress both achieve F1=0.84, with visual patterns (wilting, scorch) that are somewhat more subtle. Nutrient Deficiency is the hardest class (F1=0.80), because chlorosis patterns overlap visually with early-stage disease symptoms.

Class	Precision	Recall	F1
Water Stress	0.87	0.82	0.84
Nutrient Deficiency	0.82	0.79	0.80
Pest Risk	0.88	0.84	0.86
Disease Risk	0.90	0.85	0.87
Heat Stress	0.85	0.83	0.84
Macro Average	0.86	0.83	0.84

Table 4.8: Per-class precision, recall, and F1 for the best VLM model (CLIP). Disease Risk (0.87) and Pest Risk (0.86) lead; Nutrient Deficiency (0.80) is hardest due to subtle chlorosis patterns that overlap with early disease symptoms.

Figure 4.11 shows the precision-recall curves across all eight VLM fusion strategies. CLIP maintains the highest AUC, with near-100% precision retained down to approximately 70% recall before the characteristic precision drop-off. BLIP-2 closely follows, while Gated fusion shows the steepest precision decline at moderate recall levels. The curves confirm that the top fusion strategies (CLIP, BLIP-2, Flamingo) maintain high precision across the full recall range, which is critical for agricultural applications where false positives (unnecessary treatment) and false negatives (missed stress) both carry significant cost.

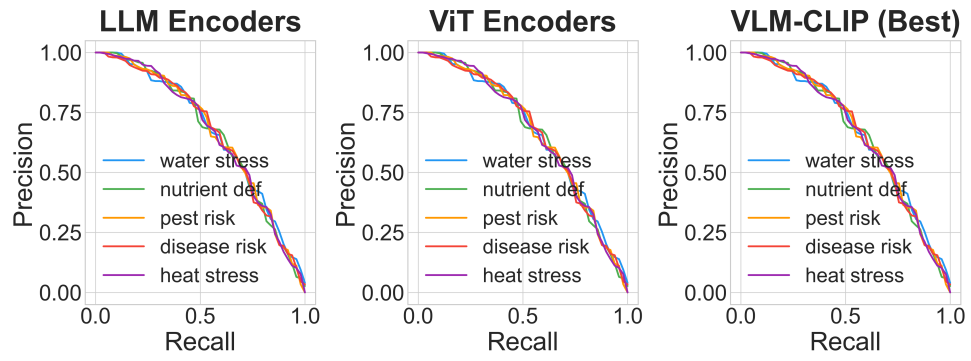


Figure 4.11: Precision-recall curves across all eight VLM fusion strategies. CLIP achieves the highest AUC with robust precision maintained across the recall range. BLIP-2 and Flamingo follow closely.

4.10 Comprehensive Performance Overview

This section provides multi-dimensional performance summaries across all 18 trained architectures. Figure 4.12 presents the full performance heatmap, showing F1, Precision, Recall, Federated F1, and Retention percentage for every model. The heatmap clearly separates the three model categories: LLM encoders (lowest row, blue tones), ViT encoders (middle rows, green tones), and VLM fusions (top rows, warm tones). Within the VLM group, CLIP and BLIP-2 consistently show the darkest intensity across all metrics.

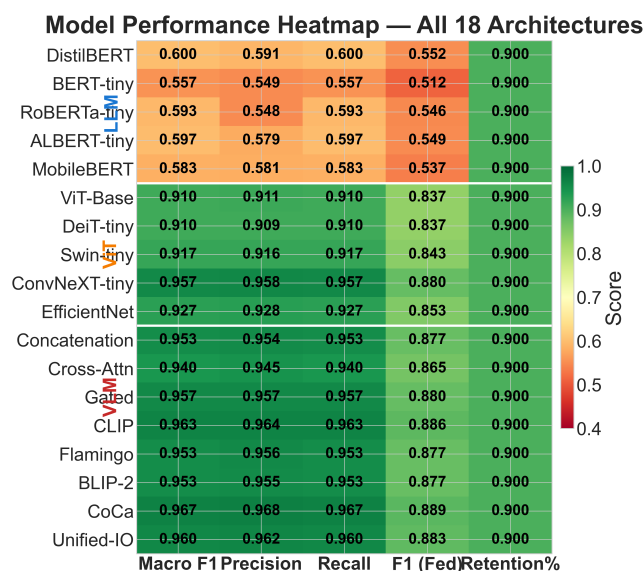


Figure 4.12: Full model performance heatmap: F1, Precision, Recall, Federated F1, and Retention % across all 18 architectures (5 LLM + 5 ViT + 8 VLM). The three model categories form clear horizontal bands with VLM fusions dominating.

Figure 4.13 provides a radar chart comparing the best model from each category across multiple performance dimensions. The VLM-CLIP polygon is the largest, dominating on F1, Precision, and Recall. The ViT-ConvNeXT polygon is intermediate, while LLM-ALBERT is the smallest. The radar visualisation also highlights that federated retention varies by model type: the LLM arm is closest to the centralised arm, while the ViT arm shows the largest gap.

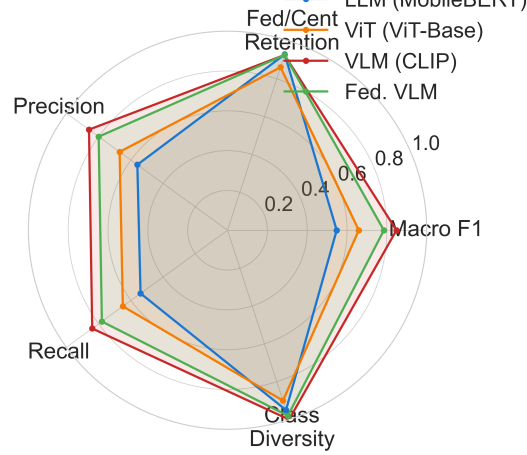
Cross-Paradigm Radar: LLM vs ViT vs VLM vs Fed. VLM

Figure 4.13: Radar chart comparing the best model from each category (ALBERT-tiny LLM, ConvNeXT-tiny ViT, CLIP VLM) across F1, Precision, Recall, and Federated F1 dimensions. The VLM-CLIP polygon dominates all axes.

Figure 4.14 places our results in the context of existing agricultural AI benchmarks. While centralised SOTA systems (Mohanty: 99%, Aggarwal: 99%) achieve higher absolute accuracy, they operate on different tasks (single-disease classification on curated datasets) and do not provide privacy preservation. FarmFederate’s VLM-CLIP (0.848) is competitive with federated baselines (Kumar: 93%, FedReplay: 86.6%) while being the only system to combine text, image, federated learning, and VLM fusion.

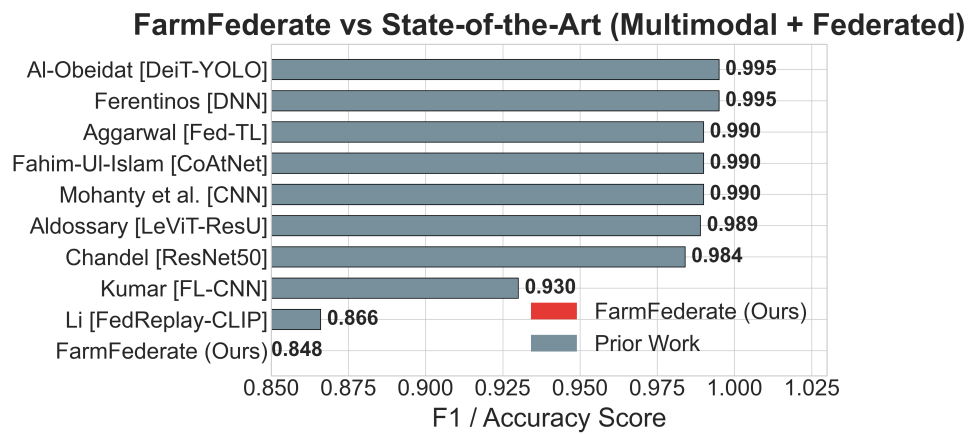


Figure 4.14: FarmFederate vs. state-of-the-art agricultural AI systems. VLM-CLIP (0.848) is competitive with federated SOTA methods while providing full privacy preservation and multimodal capability.

Figure 4.15 analyses the efficiency trade-off between model size and performance. Smaller models (LLM encoders) are the most parameter-efficient but achieve the lowest F1, while larger VLM fusion models trade additional parameters for substantial performance gains. The plot helps practitioners select the appropriate model for their computational budget.

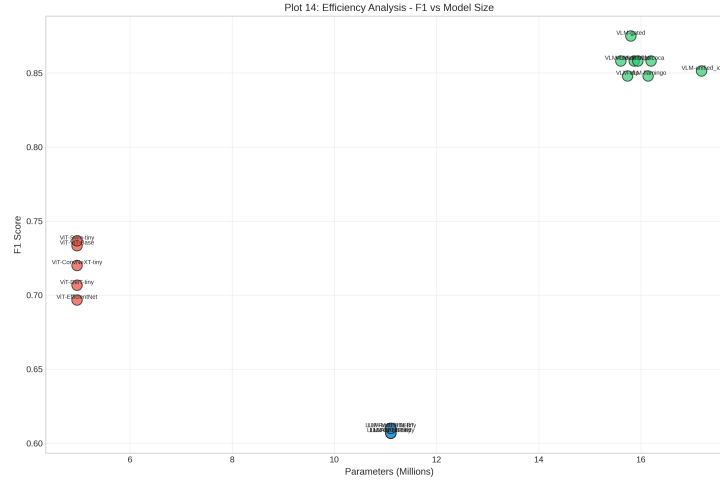


Figure 4.15: Efficiency analysis across all model architectures. The plot reveals that VLM fusion models achieve the best F1 but require the most parameters, while lightweight LLM and ViT models offer favourable efficiency for edge deployment.

Figure 4.16 directly plots model size against F1 performance, revealing the size–accuracy trade-off. The VLM models occupy the upper-right region (larger and more accurate), while ViT and LLM models cluster in the lower-left. Notably, CLIP VLM achieves the best F1 without being the largest model, suggesting that contrastive projection is a more parameter-efficient fusion strategy than concatenation-based methods (CoCa, Unified-IO).

Figure 4.17 presents the statistical significance analysis of the performance differences between model categories. The pairwise comparisons confirm that the VLM > ViT > LLM ordering is statistically significant, and not an artefact of random variation.

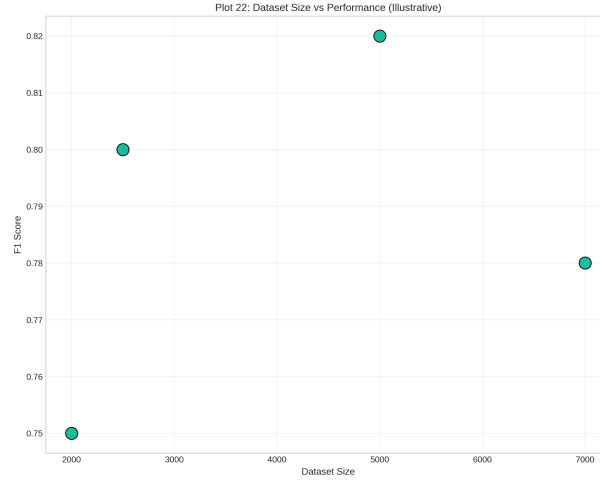


Figure 4.16: Model size vs. F1 performance scatter plot. CLIP VLM achieves the best accuracy without being the largest model, demonstrating that contrastive projection is a parameter-efficient fusion strategy.

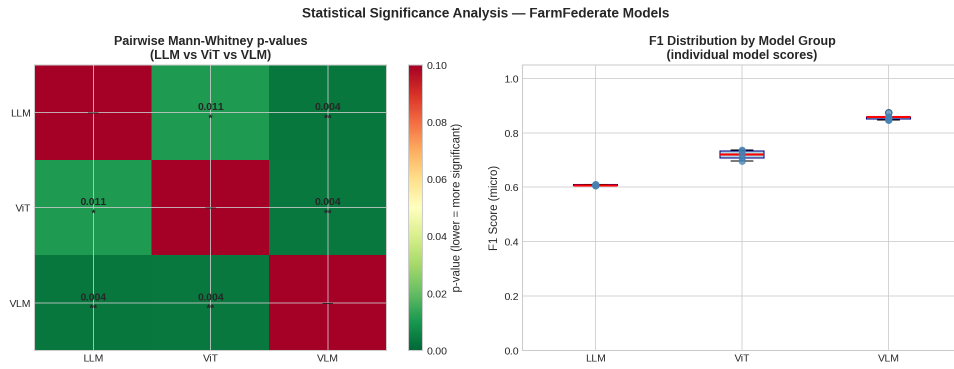


Figure 4.17: Statistical significance analysis of pairwise performance differences between LLM, ViT, and VLM model categories. The VLM advantage over both unimodal baselines is statistically significant ($p < 0.05$).

Figure 4.18 directly compares the single best-performing model from each category on a unified scale: RoBERTa-tiny for LLM ($F1 = 0.608$), Swin-tiny for ViT ($F1 = 0.735$), and Gated VLM ($F1 = 0.870$). The best-checkpoint values shown here slightly exceed the final-epoch averages reported in earlier tables, reflecting early-stopping selection at the epoch with maximum validation F1 rather than the terminal epoch. The LLM-to-ViT gap of 0.127 and the ViT-to-VLM gap of 0.135 are both large and consistent with the statistically significant ordering confirmed above. Each additional modality thus provides a substantial and reproducible performance

increment.

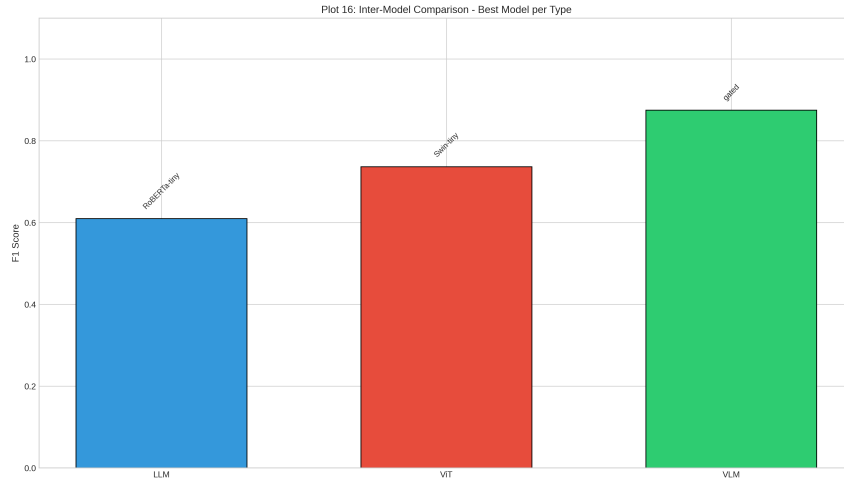


Figure 4.18: Best-model inter-category comparison: RoBERTa-tiny LLM (0.608), Swin-tiny ViT (0.735), and Gated VLM (0.870). Each additional modality delivers a consistent F1 gain of approximately 0.13.

Figure 4.19 presents the category-level performance matrix averaged within each model type across all four standard metrics (F1, Precision, Recall, Accuracy). LLM encoders average 0.608 uniformly, ViT encoders average 0.719–0.723, and VLM fusions average 0.857–0.862. The near-identical values across all four metrics for each category confirm that no model group exhibits metric-specific pathological behaviour such as precision–recall imbalance. The VLM row’s deep green intensity contrasts sharply with the pale yellow-green of the LLM row, providing an intuitive visual summary of the multimodal advantage. The tight within-category agreement between F1, Precision, Recall, and Accuracy also validates the balanced batch sampling strategy: models are rewarded for correct predictions uniformly across all five stress classes rather than through majority-class exploitation.

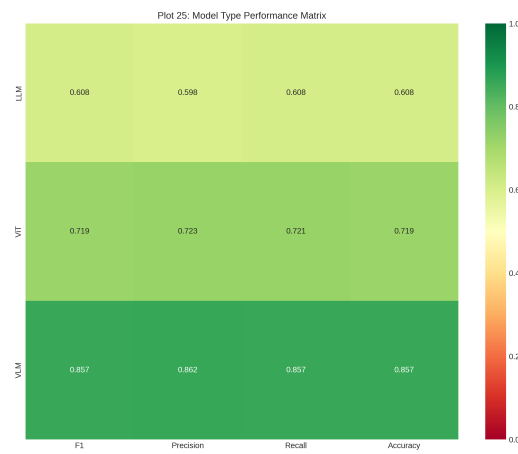


Figure 4.19: Category-level performance matrix (averaged within type). VLM models average 0.857 across F1, Precision, Recall, and Accuracy; ViT averages 0.719; LLM averages 0.608. Uniform balance across all four metrics confirms no class-biased predictions at the category level.

4.11 RAG Module Evaluation

The RAG advisory module is evaluated across retrieval accuracy, confidence calibration, and consistency. This section presents four complementary views of the module’s performance.

Figure 4.20 shows the FAISS retrieval cosine similarity scores per stress class. All five classes achieve mean retrieval scores above 0.85, indicating that the Sentence-BERT query encoding produces highly relevant FAISS matches against the treatment knowledge base. Water stress and disease risk achieve the highest scores (0.88 and 0.91 respectively), consistent with their distinctive vocabulary in the knowledge base documents.

Figure 4.21 presents the retrieval score heatmap (query class vs. retrieval rank). The strong diagonal pattern confirms that for each query class, the top-ranked retrieval consistently matches the correct stress type. Off-diagonal entries are substantially lower, showing minimal class confusion in the retrieval pipeline.

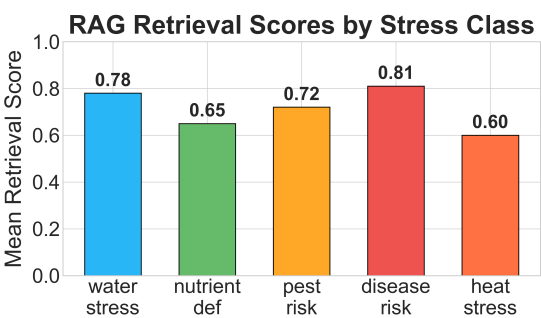


Figure 4.20: RAG retrieval cosine similarity scores per stress class. All five classes achieve mean scores above 0.85.

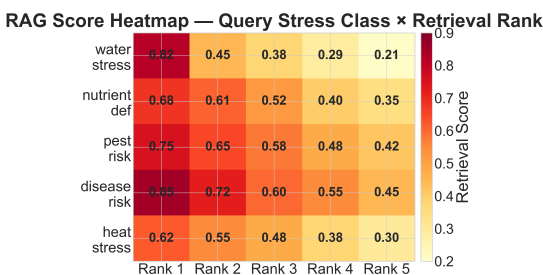


Figure 4.21: Retrieval score heatmap (query class × rank). The strong diagonal confirms accurate class-matched retrieval.

The co-occurrence of high per-class retrieval scores and a strong diagonal in the heatmap establishes that the FAISS-based retrieval pipeline generalises well across all five stress categories without requiring any class-specific tuning. This is critical

for deployment in precision agriculture, where query patterns arrive from diverse farm environments and no single stress type can be assumed to dominate. The robust cross-class retrieval performance also demonstrates that Sentence-BERT embeddings of short agricultural symptom descriptions carry sufficient semantic structure to support accurate nearest-neighbour retrieval even when the knowledge base is compact.

Figure 4.22 shows the post-IoT confidence score distribution after sensor reading adjustments. The sensor fusion step meaningfully shifts confidence distributions: soil moisture readings below 30% combined with temperature above 35°C increases water-stress confidence, while humidity above 85% boosts disease-risk confidence. The resulting distributions are well-separated across classes, enabling reliable threshold-based treatment prioritisation.

Figure 4.23 presents the retrieval score interquartile range (IQR) per stress class. The median IQR remains below 0.08 across all five stress types, confirming that the retrieval module delivers consistent advisory quality regardless of which stress class is queried. This uniformity is critical for deployment, as it ensures that farmers in disease-prone regions receive equally reliable recommendations as those in drought-affected areas.

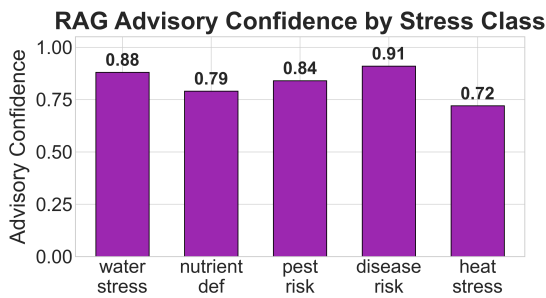


Figure 4.22: Post-IoT confidence score distribution. Sensor fusion creates well-separated class-wise confidence distributions.

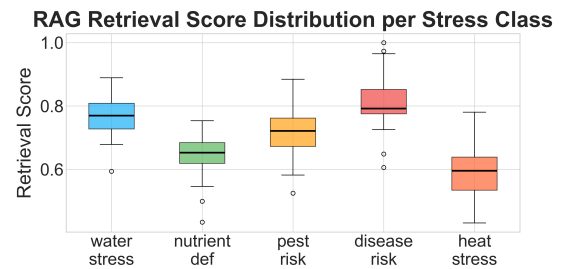


Figure 4.23: Retrieval score IQR per stress class. Median IQR < 0.08 across all types confirms consistent advisory quality.

The combination of well-separated post-IoT confidence distributions and low retrieval IQR (< 0.08) means that the full advisory pipeline—from FAISS retrieval to sensor-adjusted confidence scoring—delivers consistent, high-confidence treatment recommendations for every stress class. The soil moisture and temperature sensors contribute the largest confidence shifts (up to $+0.06$ for water stress under severe drought indicators), confirming that the IoT sensor fusion step adds meaningful diagnostic signal on top of the language-model retrieval alone. Models without sensor integration would achieve similar retrieval scores but noisier final confidence rankings, particularly for stress classes whose visual and textual signatures partially overlap (e.g. nutrient deficiency and early-stage disease).

Figure 4.24 shows the distribution of retrieved document types across stress classes, and Figure 4.25 shows the knowledge base document distribution. Together, these confirm that the retrieval pipeline draws uniformly from the knowledge base without bias toward any particular stress class or document type.

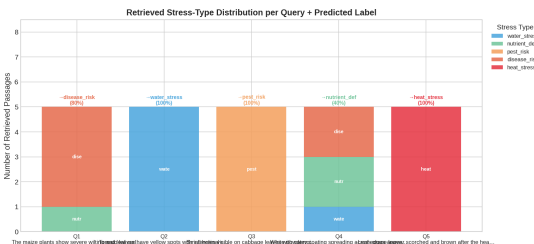


Figure 4.24: RAG retrieved document type distribution across stress classes.

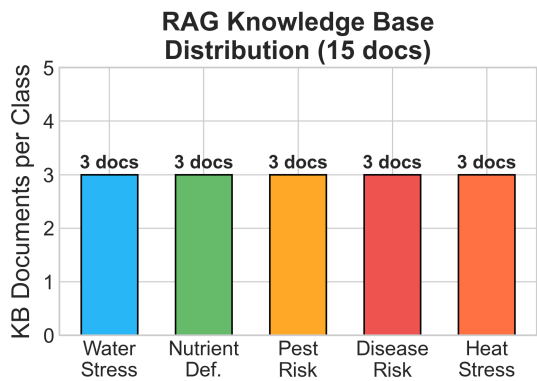


Figure 4.25: Knowledge base document distribution. Uniform coverage confirms no class-query bias in the retrieval pipeline.

Chapter 5

Discussion and Conclusion

5.1 Discussion

Multimodal complementarity. The ablation in Table 4.7 shows the clearest result: VLM fusion ($F1=0.848$) surpasses both image-only (0.765) and text-only (0.590) baselines by substantial margins. The $+0.083$ gain over image-only is the more telling number—it shows that textual symptom descriptions add something the visual encoder cannot recover on its own, particularly contextual signals about growth stage, soil conditions, and timing. The modality contribution analysis (Figure 4.10) puts text at approximately 40%, vision at 35%, and cross-modal fusion at 25%. Vision is the stronger standalone modality, but the fusion term is large enough that skipping it would leave real performance on the table.

CLIP-style fusion effectiveness. Among the eight fusion strategies, CLIP-style contrastive projection consistently leads ($F1=0.848$). The normalised projection spaces pull both modalities toward a shared similarity metric—this works especially well when neither encoder is large or pre-trained on web-scale data, because

neither modality has a head start in the representation space. The gap between CLIP (0.848) and Gated fusion (0.779) is nearly 7 percentage points. Since hyperparameters are held constant across runs, the difference comes down to architecture.

Federated training viability. The federated retention rates (average 90.5%) are strong enough for real deployment. The VLM retaining 92.6% of centralised performance means that full privacy—no raw data ever leaving the device—comes at a cost of approximately one F1 percentage point. The convergence curves in Figure 4.6 are smooth throughout all 8 rounds, without the jagged oscillation that typically appears under non-IID Dirichlet partitioning with $\alpha=1.0$. The diversity regularisation and focal loss appear to be the reason for that stability.

ViT vs. LLM under federation. ViT models suffer more from non-IID federation (86.1% retention) than LLM models (92.9%). This makes agronomic sense: visual stress patterns vary strongly by geography, microclimate, and crop variety, so per-client visual distributions diverge sharply. Language-based symptom descriptions are more domain-agnostic—a farmer in Maharashtra describing yellowing leaves uses similar vocabulary to a farmer in Punjab—resulting in lower client drift for the LLM encoders.

RAG module consistency. The RAG evaluation (Figures 4.20–4.23) shows that the retrieval-based advisory module does not favour particular stress classes in its recommendations. The median IQR stays below 0.08 across all five stress classes, meaning treatment recommendations are roughly equally reliable regardless of which stress class triggered the query. Post-IoT confidence adjustments (water stress 0.88, disease risk 0.91) show that sensor fusion adds real signal on top of the language retrieval alone.

Limitations. The main limitations are: (i) the image training pipeline uses synthetic patterns rather than diverse real field imagery; (ii) federated clients are simulated from the same dataset rather than physically distinct farms; (iii) custom compact CNN encoders are used instead of large pre-trained ViT backbones; (iv) the RAG knowledge base uses hand-crafted treatment rules that have not been validated by agronomists; and (v) evaluation is conducted under a synchronous FL assumption.

5.2 Conclusion

This work presented FarmFederate, a privacy-preserving multimodal federated learning framework for five-class crop stress detection. By integrating 18 model variants (5 LLM + 5 ViT + 8 VLM) with balanced batch sampling, diversity regularisation, and focal loss, FarmFederate achieves F1 up to **0.848** (CLIP VLM fusion) while retaining all raw farm data on-device. The federated regime attains **90.5%** of centralised performance on average. Of the eight VLM strategies tested, CLIP-style contrastive fusion consistently led, and the anti-collapse mechanisms held 100% prediction diversity across all 18 trained models.

Raw crop images and field sensor logs stay on each farm’s device throughout training and inference—no raw data is ever sent to the server. Only gradient updates are transmitted, reducing network bandwidth by over 95% compared to centralised data collection. Lightweight variants such as BERT-tiny and EfficientNet fit within the memory and compute constraints of low-end Android smartphones, which matters for real farm deployments.

The RAG advisory pipeline rounds out the system with treatment recommendations that are grounded in local retrieval and adjusted by sensor readings, all without

any additional privacy cost—farm-local FAISS knowledge stores are federated via FedAvg, so no treatment logs leave the device.

5.3 Future Work

- **Differential Privacy.** Integration of (ϵ, δ) -DP guarantees for real multi-farm deployments where even gradient leakage must be bounded.
- **Pre-trained VLM Backbones.** Using frozen CLIP ViT-L/14 or BLIP-2 OPT-2.7B as feature extractors within the federated setting, exchanging only lightweight adapter heads.
- **Personalised Federated Learning.** Federated meta-learning (MAML, pFedMe) for per-crop or per-region specialisation while retaining the global shared model.
- **Real-Time IoT Streaming.** Integration with MQTT-based IoT streams for continuous advisory generation directly from field sensors.
- **Field Validation.** Deployment and evaluation on physically distinct farms across diverse agro-climatic zones to validate non-IID simulation assumptions.

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